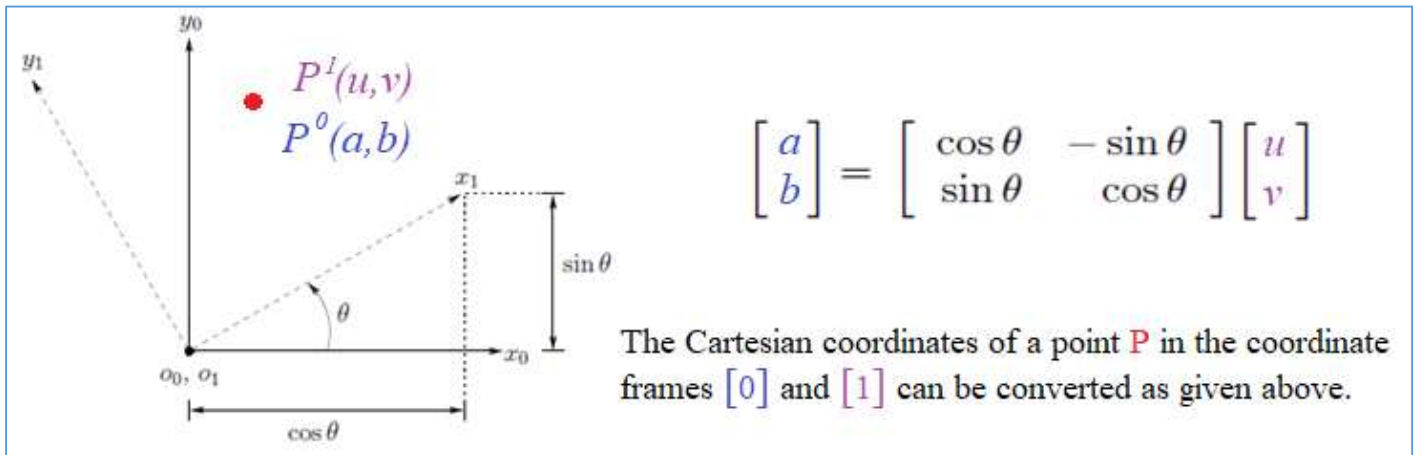


Quiz #1



Write a Python program to

- (1) **Read** from the file “**data1.txt**” of the coordinates of points defined in the coordinate frame **[1]** and stores them in a “N x 2” array. (20%)
- (2) **Read** from the console of an rotation angle θ (in degrees) (10%)
- (3) **Create a function** that takes an angle (in degrees) and the “N x 2” coordinates array in step (1) as input arguments, and returns the converted coordinates in frame **[0]** in another “N x 2” array. (40%)
- (4) **Sort** the points according to their converted X-coordinate (in frame **[0]**) in the **ascending** order (small to large). If using numpy array, sorting can be done as below. (10%)

```
# Example - Sort 2D NumPy array numpyArr by 1st Column
columnIndex = 0
sortedArr = numpyArr[numpyArr[:, columnIndex].argsort()]
```

- (5) **Write** the coordinates of **sorted** points in step (4) to a CSV file “**frame0.csv**” in which one point per line with 3 digits after decimal point for a float number is observed. A sample is given here. (20%)

```
-5.849,0.330
-2.500,4.330
0.866,0.500
1.232,1.866
1.598,3.232
1.964,4.598
```

execution

The screenshot shows a Python 3.8.3 Shell window with the following output:

```
Python 3.8.3 (default, Jul 2 2020, 17:30:36) [MSC v.1916 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
===== RESTART: C:\Python38\ml\quiz1.py =====
Enter the rotation angle in degrees > 30
6 points are converted.
>>>
```

Below the shell window are two Notepad windows. The left window, titled 'data1.txt', contains the following data:

```
[1.0 0.0]
[2.0 1.0]
[3.0 2.0]
[4.0 3.0]
[0.0 5.0]
[-4.9 3.21]
```

The right window, titled 'frame0.csv', contains the following data:

```
-5.849,0.330
-2.500,4.330
0.866,0.500
1.232,1.866
1.598,3.232
1.964,4.598
```

In the foreground, an Excel window titled 'frame0.csv - Excel' displays the data from the 'frame0.csv' file. The data is organized into columns A and B, with rows 1 through 6. The values in column A are -5.849, -2.5, 0.866, 1.232, 1.598, and 1.964. The values in column B are 0.33, 4.33, 0.5, 1.866, 3.232, and 4.598.

	A	B
1	-5.849	0.33
2	-2.5	4.33
3	0.866	0.5
4	1.232	1.866
5	1.598	3.232
6	1.964	4.598